

View Reviews

Paper ID

262

Paper Title

Predictive KV Cache Amortization for Edge Language Model Inference on Mobile Devices

Track Name

Track 2: Hardware Systems and Architectures for Edge AI

Reviewer #1

Questions**1. Overall Merit**

Accept

2. Rate the Novelty of the Idea (1: Not Novel at all, 5: Extremely Novel)

4

3. Rate the Technical Depth of the Paper (1: Extremely Weak, 5: Very Strong)

3

4. Rate the Readability and of the Paper (1: Very Hard to Read, 5: Thoroughly Enjoyable)

3

7. Strengths (Bullet Points)

1. The idea to use sensor outputs for context generation.
2. End to end implementation of the context generation and updation policies.

8. Weaknesses (Bullet Points)

1. There is no comparison of this work with the RAGCache [1] or the SGLang[2]. The latter reuses KVCache with LRU policy and the former proposes a better performing algorithm than LRU for KV Caches. RAGCache also performs prefill of the KVCache.
2. The abstract and introduction sections of the paper hastily describe technical details without setting necessary context for the readers. Probably this is because of the heavy reliability on AI tools in writing the manuscript.

[1] Chao Jin, Zili Zhang, Xuanlin Jiang, Fangyue Liu, Shufan Liu, Xuanzhe Liu, and Xin Jin. 2025. RAGCache: Efficient Knowledge Caching for Retrieval-Augmented Generation. ACM Trans. Comput. Syst. 44, 1, Article 2 (February 2026), 27 pages. <https://doi.org/10.1145/3768628>

[2] Lianmin Zheng, Liangsheng Yin, Zhiqiang Xie, Jeff Huang, Chuyue Sun, Cody Hao Yu, Shiyi Cao, Christos Kozyrakis, Ion Stoica, Joseph E. Gonzalez, Clark Barrett, and

Ying Sheng. 2023. Efficiently programming large language models using slang.
arXiv:2312.07104

9. Specific Comments to the Authors (visible after Author Notification)

1. I think the novelty of this work was to employ the sensor data into the context. The experiments demonstrating the advantage of this approach are not present.

Reviewer #3

Questions

1. Overall Merit

Weak Accept

2. Rate the Novelty of the Idea (1: Not Novel at all, 5: Extremely Novel)

3

3. Rate the Technical Depth of the Paper (1: Extremely Weak, 5: Very Strong)

4

4. Rate the Readability and of the Paper (1: Very Hard to Read, 5: Thoroughly Enjoyable)

4

7. Strengths (Bullet Points)

Complete implementation on real hardware, not a simulation, with credible native std::chrono instrumentation (isolated llama_decode, monotonic clock, a parser that aborts on trial mismatch) and reported variances.

The novel framing (sensor-driven idle-time priming for a single-user device) is well-positioned against server-side prefix reuse, which correctly does not transfer with no shared request stream.

Real engineering care: scalar-only JNI boundary, reference-counted eviction, the runInference vs resumeInference position-ID distinction, atomic rename for crash consistency.

Unusually honest limitations (single device/model, synthetic fusion data, no power telemetry, no non-predictive baseline) and a specific AI-use disclosure.

8. Weaknesses (Bullet Points)

Headline compares unequal contexts. CAP-KVC caches about 50-token prompts but the 40.78x/4.24x are stated against RAG at N about 500, conflating the caching benefit with a context-length difference. The authors flag this in one line, but the abstract, contributions, and conclusion still lead with the raw multipliers.

No non-predictive cache baseline. Without an oracle-hit or always-resident variant, the speedup is attributable to KV save/restore (a llama.cpp capability), and the paper's distinctive claim (that sensor prediction is what makes this work) is unmeasured.

The prediction layer is never evaluated. The scoring function and drift detector are described in detail, but there is no hit rate, cold-miss rate, or sensor-trace dataset.

Every latency number assumes a cache hit; how often the Tier-1 fallback fires is untested.

End-to-end win is smaller than the prefill multiplier implies (generation dominates at 3500 to 5400 ms), and the N about 50 end-to-end result is non-significant ($p = 0.209$).

9. Specific Comments to the Authors (visible after Author Notification)

Strong systems paper; my concerns are framing and two missing experiments, not correctness.

Attach the context-inequality caveat to the headline everywhere, not just the one "Important limitation" line. State that the result shows length-independent restoration and that the multiplier grows with baseline context length by construction.

Add a non-predictive cache baseline (oracle-hit or always-resident). This is the experiment that separates the KV-restore benefit from your actual novelty, the sensor prediction.

Evaluate the prediction layer, at least offline: context hit rate, cold-miss rate, and Tier-1 fallback frequency on sensor-context traces. This is the load-bearing assumption of the system.

Consider cutting or relocating the multimodal fusion study. It uses synthetic data, its pre-registered hypothesis failed, and the 42.4% LOSO EMG with 0% reject recall raises orthogonal concerns.

Minor: quantify battery cost of the 60-second sweep and 15-minute poll; add one sentence on the threat model for the unencrypted cache; ref [22] is a weak anchor for on-device prefill scaling.

Reviewer #4

Questions

1. Overall Merit

Weak Reject

2. Rate the Novelty of the Idea (1: Not Novel at all, 5: Extremely Novel)

2

3. Rate the Technical Depth of the Paper (1: Extremely Weak, 5: Very Strong)

2

4. Rate the Readability and of the Paper (1: Very Hard to Read, 5: Thoroughly Enjoyable)

3

7. Strengths (Bullet Points)

Functioning, non-trivial mobile implementation: careful JNI boundary design (scalar-only, 9 native functions), single-engine-lock concurrency discipline across cache load and resume, atomic-rename persistence for crash safety, reference-counted LRU

eviction.

Correctly identifies that standard server-side KV-reuse techniques don't transfer to single-user edge devices lacking a shared multi-tenant request stream.

Unusually transparent limitations reporting — flags the context-richness mismatch in its own comparisons, admits the drift-detector's k value isn't supported by its offline experiment, and disclaims clinical relevance of the multimodal module.

Reasonably careful low-level latency instrumentation and appropriate statistical testing methodology (Welch's t -test, effect sizes, explicit normality caveats).

8. Weaknesses (Bullet Points)

The paper's central premise is untested. No measurement is given of how often the sensor-based predictor actually selects the correct context in realistic conditions — this is the one number that would determine whether the system helps real users, and it's absent. The reported speedups only show that restoring a correctly-precomputed cache is fast, which does not require this paper to establish.

No accounting for the cost side of prediction: wasted idle-time compute and battery drain when predictions are wrong, cold-miss frequency, or fallback-path latency in practice.

Core drift-detection parameter ($k=3$) is deployed despite the authors' own offline evaluation favoring a different value ($k=1$); the methodological mismatch used to justify this discrepancy is acknowledged but leaves the deployed system's key stability parameter without real support.

Evaluation scope is narrow: one device (OnePlus 11R), one small model (Qwen2.5-0.5B Q4_K_M), $n=30$ trials per condition — insufficient to support generalizable systems claims.

No thermal, CPU-frequency, or power/battery measurements, despite the architecture's entire value proposition resting on exploiting "free" idle-time compute — battery cost of always-on sensing (60s sweeps, 15-min WorkManager polling) is explicitly left uncharacterized.

Context-richness confound: CAP-KVC caches are derived from ~50-token prompts but compared against RAG_INLINE at up to 500 tokens, which the authors themselves flag as limiting the comparison's validity but do not resolve.

The multimodal intent module (sEMG+gaze) adds little evidential value — synthetic data only, weak LOSO performance (42.4% accuracy, 0% recall on one class) — and is largely disconnected from the core caching contribution, arguably diluting the paper's focus.

9. Specific Comments to the Authors (visible after Author Notification)

The engineering is solid, but the central claim is untested: nothing measures how often the sensor-based predictor actually guesses the right context, so the reported speedups only show a cache hit is fast — not that this prediction strategy helps in practice versus a simpler baseline. A small trace-based accuracy study would close this gap.

The drift-detector's $k=3$ is also unvalidated: the paper's own ablation favored $k=1$ but

tested a different (Sentence-BERT/DailyDialog) mechanism than the one deployed. Evaluating k against the actual sensor-hysteresis logic would strengthen this choice considerably.

Since the system's premise rests on idle-time compute being "free," battery/thermal cost of the always-on GPS/BLE sensing and 15-minute polling should be reported — this tradeoff matters for a device users depend on all day.

Finally, there's no adversarial threat model: GPS can be spoofed via mock-location tooling and BLE proximity via cheap beacon impersonation, with no defense against readings that are deliberately falsified rather than merely noisy. Combined with the paper's admission that cached files are unencrypted, and given that misattributed context (wrong location, wrong assumed companion) could shape what a non-speaking user's device says on their behalf, this deserves at least a dedicated discussion.